

# Cybersecurity Incident Report:

## Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that the udp packet was undeliverable to port 53 of the dns server. This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: "udp port 53 unreachable". The port noted in the error message is used for dns service. The most likely issue is the udp message requesting an ip address for the domain "www.yummyrecipiesforme.com" didn't go through to the dns server because no service was listening on the receiving dns port.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

The incident occurred at 13:24:32.192571. The IT team were alerted to the incident due to several customers of the client reporting that they could not access the client company website "www.yummyrecipiesforme.com" and saw the error "destination port unreachable" after waiting for the page to load. The IT department visited the website and were met with the same error message. To troubleshoot this, they used tcpdump, a network analyzer tool, and attempted to load the page again. With this, tcpdump showed that when the udp packet was sent to the dns server, icmp packets were received with the error message "udp port 53 unreachable". Port 53 is used for dns service. The likely cause of the error is there was no service listening on the receiving dns port when the udp message requested an ip address for the "www.yummyrecipiesforme.com" domain. The log shows that the icmp packets were sent an additional two times afterwards, but the same error was received both times. The issue is still being investigated and is being handled by security engineers.

